

Estimate Weight of fresh water fish of different species Using Regression Models

Abstract—Using a public fish market dataset, the study utilizes Knowledge Discovery in Databases (KDD) Methodology to develop regression model for estimating fish weight. The primary objective was to identify a model which predict most accurately by using Root Mean Square Error (RMSE) on the full dataset.

Index Terms—Regression, KDD, Fish Weight, Cube Root Transformation, Logarithm Transformation.

1. Introduction

A regression model is a statistical tool used to analyze a dataset and estimate the relationship between independent variables and a dependent target variable. This model typically uses input variables in a mathematical formula to map them to continuous numerical outcomes in order to predict trends.[1]

In this project, three regression models are used to determine the best model for predicting fish weight: multiple linear regression (MLR), a model with interaction terms, and transformed regression. MLR assumes that the relationship between the target variable and the input variables can be represented by a straight line. We used the term 'transformed regression model' to group models that apply transformations to variables to establish this linear relationship. Additionally, the interaction model incorporates the combined effects of certain variables, allowing the model to capture how variables jointly influence the target variable.

To assess the efficiency of the model, we verify the Gauss-Markov assumptions, which allows the OLS estimator to achieve the BLUE (Best Linear Unbiased Estimator) property. In addition, we use the RMSE index to measure performance; a lower RMSE indicates smaller residuals between the fitted and actual target values.

2. METHODOLOGY

To concentrate on analyzing the data and gaining a deeper understanding of the relationships between variables, the Knowledge Discovery in Databases (KDD) framework is used. This framework ensures that meaningful, accurate, and efficient insights can be extracted from the dataset. The process consists of five stages [2]:

- Selection: appropriate dataset is selected for analysis in accordance with the project objectives.
- Pre-processing: cleaning the data and handling potential issues in order to obtain a reliable dataset.
- Transformation: convert data into suitable formats for further analysis.
- Data Mining: patterns, relationships, and useful insights are discovered using analytical methods.
- Evaluation: determine model effectiveness and suitability with respect to the project objectives,

using appropriate assessment tools and performance metrics.

Because the resource that is provided with the project, the study will continue with the second step pre-processing.

2.1. Pre-processing

The dataset has 158 observations with 7 variables, there is neither empty data point nor duplicate data observations in the dataset. Each variable is described [3]:

TABLE 1: Variable Description

Var	Unit	Interpretation	Type(R)
Species	categorical		Factor
Weight	g		Numeric
Length1	cm	length from the snout to the last vertebrae	Numeric
Length2	cm	length from the snout to the fork (indent)	Numeric
Length3	cm	total length from the snout to the longest tail end	Numeric
Height	cm	height from the belly to the top (ignoring fins)	Numeric
Width	cm	width captures the thickness across the body.	Numeric

Based on a quick examination of the dataset, it contains multiple length-related variables. This suggests a high likelihood of multicollinearity among the predictors. To investigate the relationships between variables, including the target variable (y), we compute a correlation matrix for a clearer overview. Before that, the categorical variables are transformed into dummy variables using One Hot Encoding for unordered classifications.

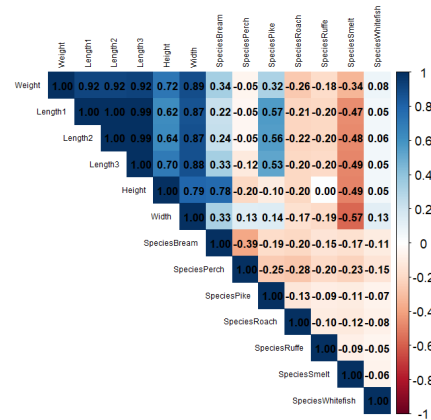


Figure 1: Correlation Matrix Heatmap

The matrix shows that the effect of independent variables upon weight is significantly obvious. And through it, we can pull out three assumptions

- A very strong correlation between three length variables, which all represent the length of the fish and are redundant. To avoid confusing and creating

'noise,' we should only keep the most important representative feature.

- Moderate correlation between Length, Height, and Width. We will evaluate these variables further to determine the optimal combination or elimination.
- The relationship between Species and Weight is split across seven different variables due to the one-hot encoding method. This decreases the visible impact of species on weight in a correlation matrix. Therefore, the matrix may not fully capture the importance of the species, which we will validate more accurately during the modeling stage.

Next step, we will look into the detail shape of each variable by using the histogram chart. Category (Species) should not using because it not the continuous numeric.

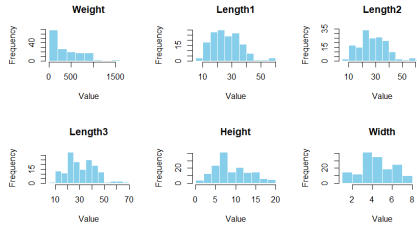


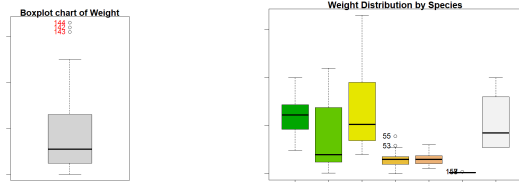
Figure 2: Histogram of each variable

It can be clearly observed that the weight variable is strongly right-skewed. In contrast, the other variables appear to have multiple peaks, properly due to differences between fish species. Nevertheless, the overall distribution of these variables still appears to roughly follow a bell-shaped pattern. To further examine, skewness and kurtosis metric are calculated for each variable.

TABLE 2: Skewness and Kurtosis of Variables

Variable	Skewness	Kurtosis
Weight	1.1201	3.9447
Length1	0.6012	3.4402
Length2	0.5525	3.4026
Length3	0.4037	3.0683
Height	0.4077	2.3817
Width	0.0130	2.4668

Weight factor actually have the strong right skewed (Skewness = 1.12, Kurtosis = 3.94) points out the potential outliers, we would like to investigate those outliers to determine if those are the important



(a) General (b) Species-specific
Figure 3: Boxplot of Weight Variable

While the general Box plot of Weight identifies rows 142, 143, 144 are global outliers, species-specific analysis provides a more comprehensive perspective. Because weight scales vary significantly across species, the outliers are simply the largest Pike fish, which more larger compare to other small fishes. Furthermore, when grouped by species, new local outliers appear (rows 53, 55 for

Roach, and 157, 158 for Smelt). After considering with their physical dimensions, we conclude these are not data entry errors but represent natural biological variance. Consequently, these data points will be retained for modeling.

Moreover, the plots illustrate a wide range in fish weights and significant positive skewness. Such distributional properties can introduce bias into the model's predictions and lead to a violation of the normality assumption of residuals. Therefore, a transformation is recommended to stabilize the variance, reduce the influence of outliers, and ultimately improve the model's predictive performance, which we worked on the transformation stage.

Scatter Plot between each variable and the target weight give the overview relationship

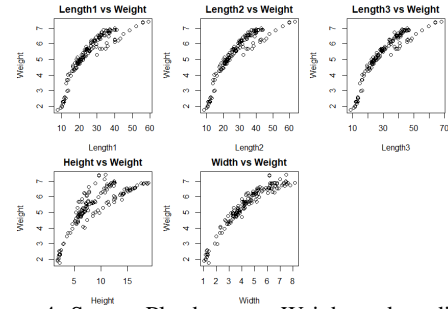


Figure 4: Scatter Plot between Weight and predictors

The plots reveal that all independent variables show a consistently curvilinear relationship with Weight, which indicate that a standard linear model would be insufficient to capture the underlying biological growth patterns.

2.2. Transformation

As the previous step, we found out the significant skewness in the Weight variable and a non-linear relationship between the target and independent variables, we will apply a transformation to the Weight variable and, based on the relationship patterns, transform the corresponding independent variables to achieve linearity and stabilize variance. The transformation will test across four forms: Square root of Weight, Cube root of Weight, Logarithm of Weight and Logarithm of all variable except Species

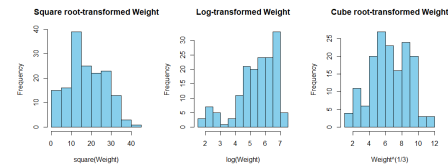
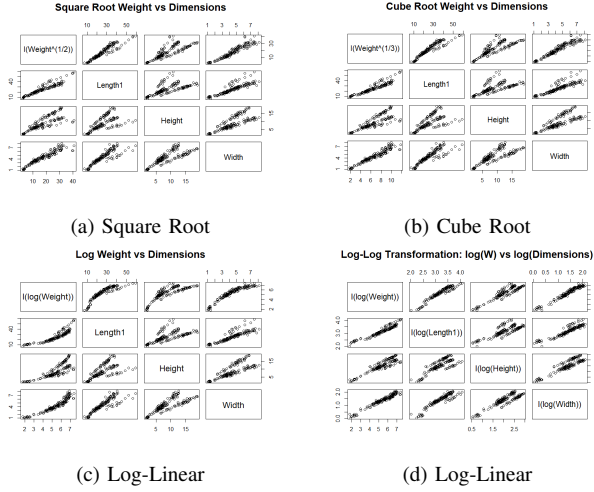


Figure 5: Histogram of Weight transformation

Form	Skewness	Kurtosis
Square Root	0.2500	2.2677
Cube Root	-0.1080	2.2137
Log	-1.0107	3.5377

Figure 6: Skewness and Kurtosis of Weight transformation

The histogram and quantitative metrics confirm that the Cube Root transformation is the most effective method (skewness: -0.10, kurtosis: 2.27). While the logarithmic form compresses the weight data set aggressively, causing a left-skew, the square root form is not strong enough to correct the original right skewness. Therefore, we will compare the relationship of each transformation with other dimensions.



These plots reveal that while the transformations all help linearize the relationship with other dimensions, the data points tend to split into two distinct directions. This indicates that the Species variable plays a crucial role in determining the weight of the fish. Additionally, the semi-log form (logarithm of weight only) still exhibits a non-linear interaction with other variables, which could be corrected by applying logarithm on all variables. Next, we will analyze the residual plots to identify which form provides the best explanatory power.

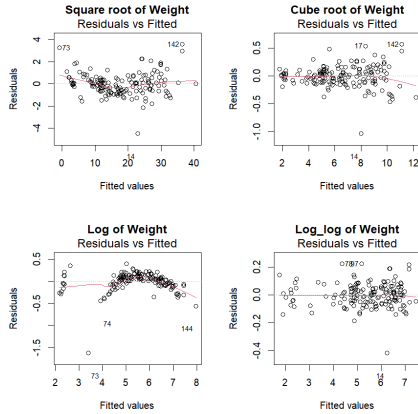


Figure 8: Residuals vs Fitted Plots using Weight transformation

The charts (Fig.8) indicate that the Square Root transformation can neither fully correct the right-skewness nor eliminate the non-linear relationship between the residuals and fitted value. While the Logarithmic model results in a slight left-skew, it is highly effective at converting curved biological growth patterns into straight lines (linearity), which is crucial for satisfying regression assumptions. Consequently, we will keep both the Cube Root and Log-Log forms for a final model comparison. The modeling step is the stage we list down all the possible model and the last evaluation stage will analyze the model performance to choose the most suitable model.

2.3. Modeling

If we use all variables for prediction, there will have a serious correlation problem, therefore, the input elimination step is compulsory and should start with take out

two of triple length variable. We have 4 models:

$$Weight = \beta_0 + \sum \gamma_i S_i + \beta_1 L_1 + \beta_2 L_2 + \beta_3 L_3 + \beta_4 H + \beta_5 W$$

$$Weight = \beta_0 + \sum \gamma_i S_i + \beta_1 L_3 + \beta_2 H + \beta_3 W$$

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The model selection was based on four statistical criteria: VIF to detect multicollinearity, ANOVA to test the significance of variable elimination, AIC to compare information efficiency among candidate models, and Pearson correlation to ensure consistency between individual features and the target variable.

Model	<i>aGVIF</i>	Anova (p_value)	AIC	Correlation
Full	>10		1883.902	
Length1	<10	0.0036	1892.054	0.915
Length2	<10	0.0566	1886.119	0.918
Length3	<10	0.0852	1885.234	0.923

TABLE 3: Length variables comparison

Results indicate that Length3 is the most effective length predictor among the individual models. While using Length1 alone leads to a significant loss of information ($p < 0.05$ in ANOVA compared to the full model), Length3 demonstrates the best balance across all performance metrics. Although the AIC for the Length3 model is slightly higher than the full model (1885.23 vs 1883.90), it significantly reduces the *aGVIF* index, eliminating severe multicollinearity. Furthermore, the ANOVA test confirms no significant loss of explanatory power compared to the full model ($p = 0.495$). Thus, Length3 is selected to ensure both model parsimony and predictive stability.

Due to the sample size of 158 observations, the data set was split into 80-20, 80% of the data for training model on historical observations and 20% were used to evaluate predictive performance on unseen data. This ratio was chosen to provide sufficient data to learn the relationships, while maintaining a large enough test sample to commence reliable evaluation metrics.

```
> table(train_data$Species)
Bream 29 Perch 43 Pike 15 Roach 14 Ruffe 8 Smelt Whitefish 11
> table(test_data$Species)
Bream 6 Perch 12 Pike 2 Roach 5 Ruffe 3 Smelt Whitefish 1
```

Moreover, the limited sample size for 7 species led to the decision to develop a global model encompassing all seven species, rather than individual models. Consequently, Species was treated as a categorical predictor within the single regression framework. The model process worked on 4 main models and 4 interaction models:

$$Weight = \beta_0 + \sum \gamma_i S_i + \beta_1 L_1 + \beta_2 L_2 + \beta_3 L_3 + \beta_4 H + \beta_5 W \quad (1)$$

$$Weight = \beta_0 + \sum \gamma_i S_i + \beta_1 L_3 + \beta_2 H + \beta_3 W \quad (2)$$

$$\ln(Weight) = \beta_0 + \sum \gamma_i S_i + \beta_1 \ln(L_3) + \beta_2 \ln(H) + \beta_3 \ln(W) \quad (3)$$

$$\sqrt[3]{Weight} = \beta_0 + \sum \gamma_i S_i + \beta_1 L_3 + \beta_2 H + \beta_3 W \quad (4)$$

2.4. Evaluation

Each model will be evaluated using BLUE properties, which find a MLR model ensure following assumptions:

Assumptions	Test Value	Pass Value
Correct functional form	Residuals-Fitted Plot	Randomness around 0 + smooth line is straight
Homoscedasticity	Scale-Location Plot	Randomness around the red vertical straight line
No autocorrelation between errors	Durbin Watson test	Approx. 2 is best value
Errors are normally distributed.	Q-Q Plot	data points locate on the 45° line
No multicollinearity between the predictors	VIF / $GVIF^{1/(2 \cdot Df)}$ ($aGVIF$)	$VIF < 10$ / $aGVIF < 2$ is acceptable
No influential data points	Residuals-Leverage plot	No point crossing the red dashed lines (Cook's Distance)

TABLE 4: Gauss Markov assumption

While the Gauss-Markov (BLUE) framework ensures our model is the most effective and unbiased linear estimator, Root Mean Square Error (RMSE) metric was used to evaluate predictive accuracy. By calculating the average error across both the test set and the full dataset, we select the final model which has the smallest RMSE .

3. Model and Evaluation

3.1. Model 1: Full model

Variable	Estimate	Pr(> t)	Signif.	$aGVIF$
Intercept	-873.988	1.84e-08	***	
SpeciesPerch	113.618	0.40917	.	
SpeciesPike	-252.731	0.09514	.	
SpeciesRoach	92.409	0.37533	.	
SpeciesRuffe	155.903	0.08126	.	
SpeciesSmelt	394.469	0.00442	**	
SpeciesWhitefish	91.227	0.41088	.	1.8
Length1	-56.024	0.17206	.	48.32
Length2	75.174	0.14629	.	65.2
Length3	17.260	0.60434	.	45.7
Height	7.794	0.60428	.	7.44
Width	-15.537	0.57635	.	5.45

Std. Error Residuals	Adjusted R-squared	Sig. Change	F	Durbin-Watson	RMSE
97.19	0.9302	0.000	1.88		85.61

TABLE 5: Model 1 - Full model outputs

The model explains over 90% (Adjusted $R^2 = 0.93$) of the variance in Weight, with a highly significant overall p-value (< 0.001), which confirms that the independent variables effectively predict the dependent variable. However, based on individual p-values, only SpeciesRuffe and Intercept appear to have a significant impact, which seems not reasonable. This is misleading due to the severe multicollinearity between the three length variables and the moderate correlation with Height and Width. Consequently, this violates the assumption of No Multicollinearity, making the individual p-values unreliable. The model is 'confused' and unable to isolate the true effect of each variable. For the next model, we should eliminate the redundant variables to slowly identify the impact of each independent variables upon weight

The diagnostic plots (Fig.9) indicate that the model violates the assumptions of linearity, homoscedasticity, and normality of errors. Specifically, the first and third

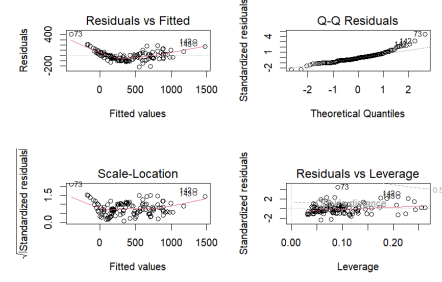


Figure 9: Model 1 - Full Model diagnostic plots

charts reveal a non-linear relationship between residuals and fitted values, as evidenced by the distinct curvilinear pattern in the residual distribution.

3.2. Model 2: Redundant variables elimination

The modelling step proved that the length3 is the most effective variable, therefore, this model will remove length1 and length2 dimensions. The output as below:

Variable	Estimate	Pr(> t)	Signif.	$aGVIF$
Intercept	-908.80	3.14e-16	***	
SpeciesPerch	176.62	0.05334	.	
SpeciesPike	-212.97	0.10531	.	
SpeciesRoach	115.45	0.17732	.	
SpeciesRuffe	180.47	0.00185	**	
SpeciesSmelt	419.23	4.56e-05	***	
SpeciesWhitefish	136.56	0.12193	.	1.58
Length3	38.73	$< 2 \times 10^{-16}$	***	5.32
Height	9.97	0.50394	.	7.37
Width	-21.04	0.44634	.	5.41

Std. Error Residuals	Adjusted R-squared	Sig. Change	F	Durbin-Watson	RMSE
97.33	0.93	$< 2.2 \times 10^{-16}$	1.94		88.40

TABLE 6: Model 2 - Elimination model outputs

After deleting the Length1 and Length2 variables, the p-value of Length3 decreased dramatically and it became the most important predictor ($p < 0.05$). This indicates that eliminating the other length variables helped the model re-estimate the impact of each independent variable. Even though the Adjusted R^2 and F-statistic are slightly lower than the full model, removing is necessary to fix the serious multicollinearity problem. This action successfully reduced the $aGVIF$ from 65.2 down to 5.07. Therefore, we chose the model with only Length3 to ensure the results are accurate and reliable. While the p-values of Width and Height suggest that these two variables do not have a large individual impact on weight, they are not definitive in this case. We suspect that length, width, and height maintain a close relationship, not only based on the $aGVIF$ index but also for biological reasons

3.3. Model 3: Cube Root

As the Pre-prepare stage above shows that cube root transformation is the perfect method to correct skewness of weight variable. We will assess following model

$$\sqrt[3]{Weight} = \beta_0 + \beta_i S_i + \beta_1 L_3 + \beta_2 H + \beta_3 W$$

Variable	Estimate	Pr(> t)	Signif.	$aGVIF$
Intercept	-0.39	0.06		
SpeciesPerch	0.39	0.04	*	
SpeciesPike	-0.24	0.37		
SpeciesRoach	0.26	0.16		
SpeciesRuffe	0.41	0.00	***	
SpeciesSmelt	0.13	0.52		
SpeciesWhitefish	0.55	0.00	**	1.58
Length3	0.13	$< 2 \times 10^{-16}$	***	5.32
Height	0.11	0.00	***	7.37
Width	0.37	$< 8.95 \times 10^{-9}$	***	5.41

Std. Error	Adjusted	Sig.	F	Durbin-	RMSE
Residuals	R-squared	Change		Watson	
0.2089	0.9928	2.2	$\times$	1.89	43.98
		10^{-16}			

TABLE 7: Model 3 - Cube Root model outputs

The model achieves a remarkable Adjusted R^2 of 99%, indicating an excellent fit to the data. By applying the cube root transformation, all physical predictors — Length, Height, and Width — become statistically significant ($p < 0.05$). This is highly reasonable because a fish's weight is physically determined by its three-dimensional body volume.

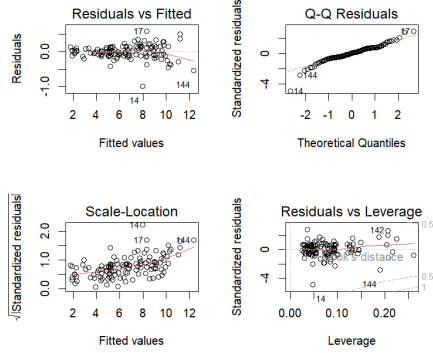


Figure 10: Model 3 - Cube Root Model diagnostic plots

Overall, the model is very effective, however, the plot charts show that it works better for smaller fish than for larger ones. In the 'Residuals vs Fitted' chart, the line drops at the end, meaning the model overestimates the weight of very large fish, which indicate the Heteroscedasticity, similar to the 'Scale-Location' chart. This suggests that while the cube-root method is great for most cases, very large fish may have different body proportions that the model cannot perfectly capture. Meanwhile, the Q-Q plot also reflects the data points at the extreme small and large ends tending to drift away from the straight line.

3.4. Model 4: Full-Log

Although the log transformation may not fully fix the skewness, it is still a powerful method to handle the wide range of fish weights and explain the curved relationship between variables. More importantly, the mathematical property $\ln(L \times H \times W) = \ln(L) + \ln(H) + \ln(W)$ helps explain how the three dimensions combine to form the weight. This approach also helps manage the mild correlation between these predictors, which is why the log-log model is examined next.

Variable	Estimate	Pr(> t)	Signif.	$aGVIF$
Intercept	-3.26	$< 2 \times 10^{-16}$	***	
SpeciesPerch	0.23	0.00	**	
SpeciesPike	0.03	0.83		
SpeciesRoach	0.14	0.05	*	
SpeciesRuffe	0.19	$< 5.82 \times 10^{-6}$	***	
SpeciesSmelt	-0.01	0.95		
SpeciesWhitefish	0.26	0.00	***	1.67
$\ln(\text{Length}3)$	1.96	$< 2 \times 10^{-16}$	***	8.90
$\ln(\text{Height})$	0.60	0.00	***	7.76
$\ln(\text{Width})$	0.51	0.00	***	11.88

Std. Error	Adjusted	Sig.	F	Durbin-	RMSE
Residuals	R-squared	Change		Watson	
0.086	0.9959	$< 2 \times 10^{-16}$	$\times$	1.75	47.51

TABLE 8: Model 4: Full Log Model Outputs

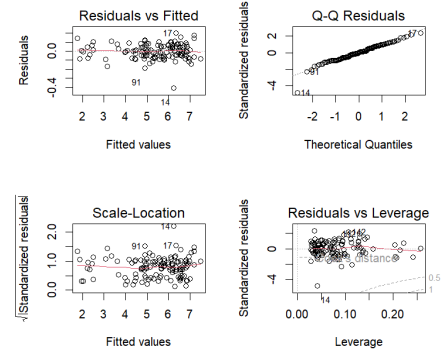


Figure 11: Model 4 - Full Log Model diagnostic plots

As expected, the log model explains the fish weight very well ($p < 0.05$). While it shows higher correlation between variables (especially Width), the log transformation successfully fixes the unequal variance and makes the errors follow a normal distribution. However, because the log scale shrinks large numbers, it can make the residual plots look better than they actually are. When we convert the results back to grams, the log model is actually less accurate than the cube-root model (RMSE = 43.98)

3.5. Model 6: Interaction Model

After evaluating the models above, we narrowed our preference down to: the Cube-root model and the Full Log-Log model. Next, we will introduce interaction terms into them to further improve their predictive power.

$$\ln(\text{Weight}) = \beta_0 + \sum \gamma_i S_i + \beta_2 \ln(L_3) + \beta_3 \ln(H/L_3) + \beta_4 \ln(W/L_3) \quad (5)$$

$$\ln(\text{Weight}) = \beta_0 + \sum \gamma_i S_i + \beta_2 \ln(L_3) + \beta_3 \ln(W * H) \quad (6)$$

$$\ln(\text{Weight}) = \beta_0 + \sum \gamma_i S_i + \beta_2 \ln(L_3) + \beta_3 \ln(H) + \beta_4 \ln(W/L_3) \quad (7)$$

$$\sqrt[3]{\text{Weight}} = \beta_0 + \sum \gamma_i S_i + \beta_{\text{inter}}(L_3 * W * H) \quad (8)$$

While the log-transformed models seem to draw a similar outputs, we select Model (5) for the next stage of analysis over Model 4. This is because Model (5) effectively reduces multicollinearity ($aGVIF < 10$) while still being very accurate. Furthermore, although the Cube-root interaction model initially showed promise, it reintroduces

severe multicollinearity issues. Therefore, we will proceed with Model 3 to be tested on the full dataset.

Index/Model	(5)	(6)	(7)	(8)
Std. Error Residuals	0.086	0.086	0.086	0.19
Adjusted R-squared	0.9959	0.9959	0.9959	0.9938
Sig. F Change	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$
Durbin-Watson	1.75	1.76	1.82	
aGVIF	< 10	< 10	> 10	> 40
RMSE	47.51	47.69	47.51	44.08

TABLE 9: Models with interaction term outputs

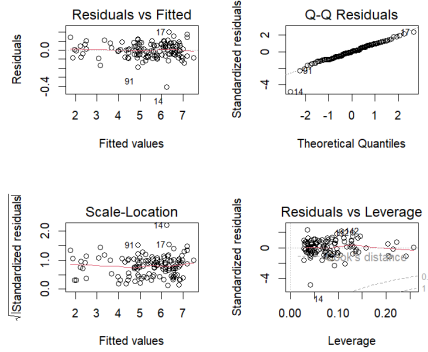


Figure 12: Model (5) - Interaction Model diagnostic plots

3.6. Conclusion

After training the model on training data set, we draw two potential model including Model (5) Full log-log with interaction model:

$$\begin{aligned} \ln(\text{Weight}) = & -3.256 + 3.067 \cdot \ln(\text{Length3}) \\ & + 0.507 \cdot \ln\left(\frac{\text{Width}}{\text{Length3}}\right) \\ & + 0.600 \cdot \ln\left(\frac{\text{Height}}{\text{Length3}}\right) + \beta_i \end{aligned}$$

Species	β_i	p-value
Bream	0	(Reference Level)
Perch	+0.233	0.0028 (**)
Pike	+0.029	0.8345
Roach	+0.138	0.0479 (*)
Ruffe	+0.190	< 0.001 (***)
Smelt	-0.007	0.9537
Whitefish	+0.258	< 0.001 (***)

TABLE 10: Coefficients of Species for Model (5)

Model (3) Cube Root Model:

$$\begin{aligned} \sqrt[3]{\text{Weight}} = & -0.3938 + 0.1329 * \text{Length3} \\ & + 0.1117 * \text{Height} + 0.3666 * \text{Width} + \beta_i \end{aligned}$$

Species	β_i	p-value
Bream	0	(Reference Level)
Perch	+0.3983	0.0425 (*)
Pike	-0.243	0.3865
Roach	+0.255	0.1643
Ruffe	+0.414	0.0008 (***)
Smelt	+0.132	0.5326
Whitefish	+0.556	0.0037 (**)

TABLE 11: Coefficients of Species for Model (3)

Model	RMSE (test)	RMSE (full)
(5)	47.51	49.77
(3)	43.98	48.25

TABLE 12: RMSE index on training, full dataset

As the objective of study to find the model that predict most accurate estimated weight, which evaluated by RMSE index, the model 3 - Cube Root model was use the calculate the future fish dataset. The estimated weight of a Perch with a height of 12.8cm, a width of 6.9cm and a total length of 41.9cm is 865.97 (R result) manual calculating as below:

$$\begin{aligned} \sqrt[3]{\text{Weight}} \approx & -0.3938 + (0.1329 \times 41.9) + (0.1117 \times 12.8) \\ & + (0.3666 \times 6.9) + 0.3983 \\ \approx & 9.5323 \end{aligned}$$

$$\Rightarrow \text{Weight} \approx 9.5323^3 \approx 866.15$$

However, to ensure the stability of the model, we should consider the BLUE properties of model:

Condition	Model (3)	Model (5)
Correct functional-form	drop down at large data points	Vertical straight line
Homoscedasticity	Random + go up at large data point	Vertical straight line
No autocorrelation	1.89	1.75
Normally distributed (Errors)	drift away at small, large data points	Straight 45° line
No multicollinearity	< 8	< 7
No influential data points	data point 144 impact strongly	No influence data point

TABLE 13: BLUE properties comparison

While the Cube Root model drew a better RMSE, the Log-Log model demonstrates the better stability and reliability by nearly satisfying all assumptions, except a high aGVIF index - a result of the inherent biological correlation between fish dimensions. Consequently, we accept a marginal trade-off in predictive precision for significantly improved model robustness, and we recommend the Full Log-Log model for all future applications.

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